



SYDNEY CAMPUS

PRT661 DATA SCIENCE PRACTICE

Assignment 01

Submitted by:

Sydney Group 3 Theme 2

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1. Project Overview and Objectives

This project uses regression-based predictive analytics on the WHO Life Expectancy dataset that provides health, economic, and social indicators for 193 countries over the years 2000 to 2015. The main aim is to develop a model that can predict a country's life expectancy based on several variables such as GDP, immunization rates, schooling, alcohol consumption, BMI, and mortality indicators.

The main objectives of our project can be described as:

- Apply data cleaning and formatting techniques to prepare the dataset for analysis.
- Perform exploratory data analysis (EDA) to discover relationships between predictors and life expectancy.
- Create and evaluate regression models such as Linear Regression, Random Forest, and Gradient Boosting to predict life expectancy.
- Assess the performance of the model with suitable measures such as RMSE, MAE, and R2.
- Demonstrate results and findings with proper plots and visualization techniques.

2. Theme Justification

This project is based on Theme 2: Predictive Analytics and Forecasting, which involves estimating life expectancy using real-world historical data. This is a project we have chosen in the context of public health. There are several factors that contribute to life expectancy. The combination of economic, social, and health-related indicators allows us to apply feature engineering methods so as to select which factors contribute most to predicting life expectancy.

As instructed in the theme, proper data handling and processing techniques will be applied. We will also build several regression models so as to compare their performance and find the best-performing one.

3. Problem Statement

Life expectancy varies greatly between different countries due to health, mortality, immunization, economic, and social issues. The dataset of WHO Life Expectancy includes all the aforementioned factors and covers 193 countries in the period between 2000 and 2015. Such factors as adult mortality, infant deaths, Hepatitis B, Polio and Diphtheria immunization, BMI, HIV/AIDS, GDP, income composition, and schooling are included.

These factors include a large amount of information on the condition of health of the people however, there are many correlations and interconnections between all these factors and life expectancy that are rather complex. As a result, it may be difficult to understand which factors are the most important to use for the prediction of life expectancy.

Thus, the goal of our project is to perform analysis using predictive analytics techniques to determine which health, economic, and social factors can predict the life expectancy with high precision. Different regression models will be compared in the process of working with WHO Life Expectancy dataset. In addition, the influence of various factors on the life expectancy prediction process will be analysed.

4. Initial Architecture Diagram

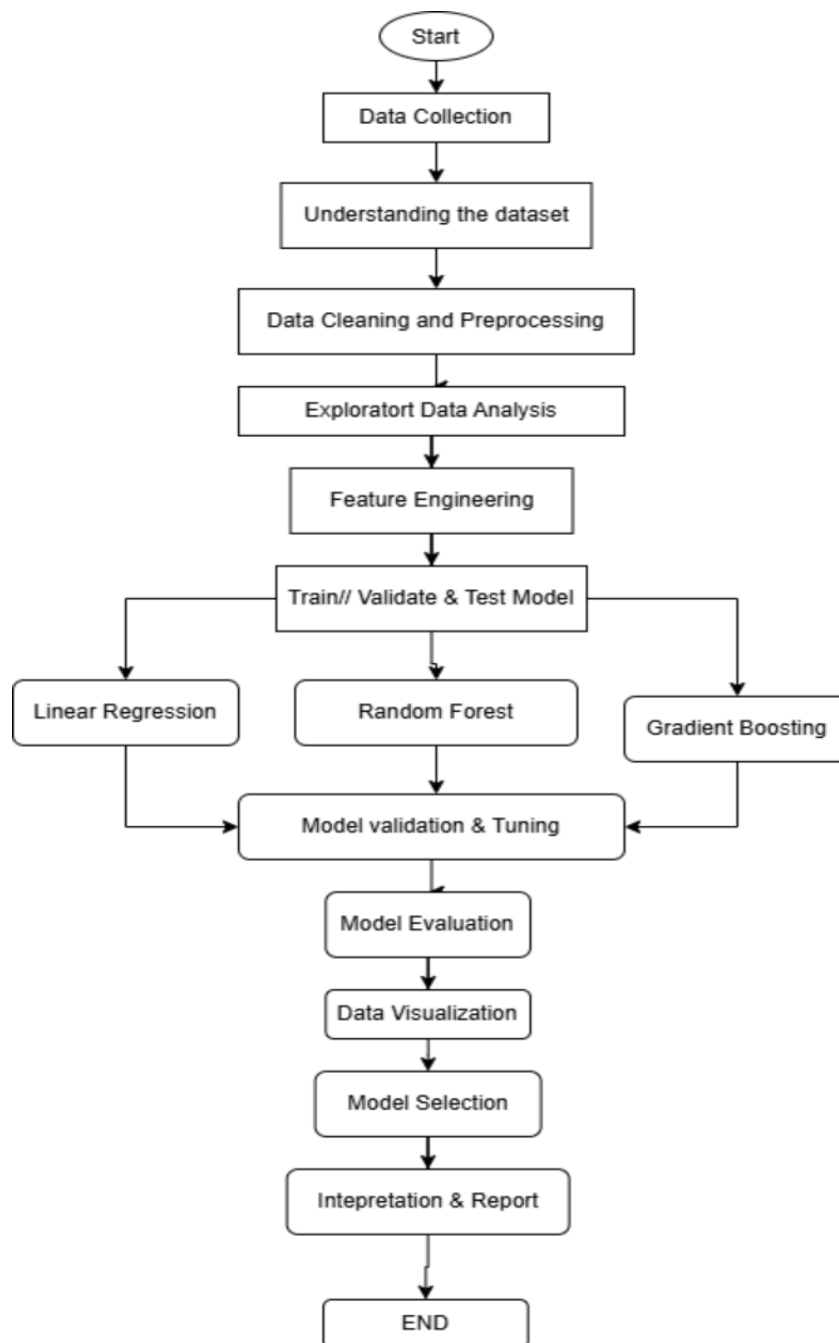


Figure 1: Predictive Analytics on Global Life Expectancy workflow

4.1 Dataset Description

Life Expectancy (WHO) has 2,938 rows and 22 columns representing various indicators related to health, economic, and social status for the period 2000-2015 for 193 countries. The dataset is publicly available on the internet from the WHO for health data analysis. We have accessed the dataset from Kaggle. The target variable is Life Expectancy. There are several features such as adult mortality, infant deaths, alcohol consumption, schooling, and income information, which are our possible predictors. The dataset contains several missing values from lesser-known countries. There is also a presence of correlation between the features, so proper data preprocessing techniques have to be applied for cleaning, transforming, and preparing the data for further analysis and visualization.

Dataset Link: <https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>

4.2 Feature Selection

Feature selection would determine which of the variables are informative for the task of estimating life expectancy. Some potential predictors for this task can be: Adult Mortality, Infant Deaths, Alcohol, BMI, Immunization, HIV/AIDS, GDP, Population, Income Composition of Resources, and Schooling. The final list of features would be chosen taking into account their relevance, quality of data, multicollinearity and impact on the performance of the model.

4.3 Model Evaluation

This project will consider the Linear Regression model and two other models from the family of tree-based models – Random Forest Regression and Gradient Boosting/XGBoost models. The Linear Regression model will serve as the baseline model, and tree-based models will be compared to see whether they offer more accurate predictions. Model evaluation will use MAE, RMSE and R^2 metrics.

5. Workflow Plan

S.N	Phase	Duration
1	Problem definition, Data Collection & Understanding	Week 1
2	Data cleaning and processing	Week 1-2
3	Exploratory Data Analysis	Week 2-3
4	Feature Engineering	Week 3
5	Train, validate and Test	Week 4
6	Model development, validation and Tuning	Week 4-5
7	Model evaluation and Selection	Week 5-6
8	Data Visualisation	Week 6
9	Interpretation and reports	Week 7
10	Project Completion	Week 7

Table 1: Workflow Plan

6. Task Allocation

Member	Role	Responsibilities
Sadan, Ashish, Pramod	Data Engineer	Data cleaning, missing value treatment (KNN/median imputation), feature encoding, outlier treatment, train/test split, GitHub setup
Sabin, Shushil	Data Analyst	EDA, correlation analysis, distribution plots, outlier detection visualisations.
Pramod, Shushil, Ashish	ML Engineer	Feature selection, model training (Linear Regression, Random Forest, XGBoost), hyperparameter tuning

Shushil, Sabin, Sadan	Model Evaluator	Cross-validation, metric computation (MAE, RMSE, R^2), model comparison and selection
Sabin, Ashish	Visualisation & Docs Lead	Results visualisation (feature importance, residual plots, predicted vs actual), report compilation and final submission

Table 2: Task Allocation

7. Initial Risk Analysis

Risk	Likelihood	Impact	Mitigation
Missing Values in Health and Economic Indicators (Hepatitis B, GDP, Population)	High	High	Identify missing values and apply appropriate imputation.
Multicollinearity (mortality variables, healthcare and education variables)	Medium	Medium	Correlation matrix, VIF analysis and feature selection.
Outliers (Population, GDP)	High	Medium	Examine distributions, proper visualization and statistical analysis.
Data quality differences between countries	Medium	High	Compare model performances across different groups, clearly document limitations in data coverage and quality.
Repeated Observations/ Overfitting	Medium	High	Cross-validation, train/test splitting.
Timeline delays	Low	Medium	Divide responsibilities among members, establish weekly milestones, maintain shared project control and progress.

Table 3: Initial Risk Analysis

8. Ethics, Privacy and Security Considerations

The dataset (WHO Life Expectancy) is publicly available and does not include any individual-level health records, however includes aggregated data for each country. So, the project does not have a direct impact on privacy of any individual as this does not include individual names, addresses or patient records. However, ethical issues remain important in the analysis and prediction of health-related outcomes.

Data Security and bias: Missing values and differences in data quality across countries can introduce bias and affect predictions.

Fair Representation: Careful interpretation of data between developed and developing countries to avoid unfair comparisons.

Data Privacy and Security: Privacy risks are low as it is publicly available.

Responsible use of predictions: Predictions will be used for academic analysis only.

References

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